

NumPy and Pandas Basics

Custom knowledge-base material for the AI & Data Science RAG Chatbot project.

1. NumPy

NumPy is a Python library for numerical computing. Its main object is the `ndarray`, a multidimensional array containing values of a compatible data type. Arrays support fast vectorized operations.

2. NumPy Operations

Common operations include creating arrays with `np.array()`, `zeros()`, `ones()`, and `arange()`. Array attributes include `shape`, `size`, and `dtype`. Broadcasting allows arithmetic between compatible array shapes without manually repeating values.

3. Pandas

Pandas is used for loading, cleaning, transforming, and analyzing tabular data. Its main structures are `Series`, a one-dimensional labeled array, and `DataFrame`, a two-dimensional table with rows and columns.

4. Loading Data

Use `pd.read_csv()` to load CSV data and `pd.read_excel()` for Excel files. Useful inspection methods include `head()`, `tail()`, `info()`, `describe()`, and `isna().sum()`.

5. Data Cleaning

Missing values can be handled by `dropna()` or `fillna()`. Duplicates can be detected with `duplicated()` and removed with `drop_duplicates()`. Data types can be changed with `astype()`.

6. Filtering and Selection

Select columns using `df['column']` or `df[['a','b']]`. Filter rows with Boolean conditions, for example `df[df['age'] > 18]`. `loc` is label-based; `iloc` is position-based.

7. Grouping and Joining

`groupby()` summarizes data by categories. Aggregation functions include `sum()`, `mean()`, `count()`, `min()`, and `max()`. `merge()` joins tables using common keys; `concat()` combines objects along an axis.

8. Exploratory Data Analysis

EDA involves understanding structure, distributions, missing values, outliers, relationships, and target variables. Visualizations such as histograms, box plots, scatter plots, and bar charts help identify patterns.